Mathieu Pont

Post-doctoral Researcher

mathieu.pont@outlook.com https://github.com/MatPont https://mathieu-pont.github.io
0000-0002-0037-0314 RPTU Kaiserslautern-Landau, Scientific Visualization Lab, Germany.

Work Experience

Oct. 2024 – Ongoing
Post-doctoral Position
RPTU Kaiserslautern-Landau, Germany (Scientific Visualization Lab).
Working with *Christoph Garth.*

Dec. 2023 – Sep. 2024
Post-doctoral Position
CNRS and Sorbonne Université (LIP6).
Working with *Julien Tierny.*

2020 (6 months)
Master 2 Research Intern
CNRS and Sorbonne Université (LIP6).
Title: *Topologically Discriminant Metric.*
Advisor: *Julien Tierny.*

2019 (3 months)
Master 1 Research Intern
Paris Descartes University (LIPADE).
Title: *Biomedical Corpus Analysis.*
Advisor: *Séverine Affeldt.*

2018 (3 months)
Bachelor Research Intern
Toulouse Paul Sabatier University (IRIT).
Title: *Comparison of Deep Reinforcement Learning methods with an existing Multi-Agent System.*
Advisors: *Frédéric Migeon and Jêrome Mengin.*

2016 (3 months)
DUT Research Intern
ISAE-Supaero.
Title: *Server Room Thermal Monitoring and Evaluation of EV3 Robotic Kit.*
Advisors: *Régine Leconte and Jean-François Dassieu.*

Education

Oct. 2020 – Nov. 2023
Ph.D. in Computer Science
CNRS and Sorbonne Université (LIP6).
Title: *Analysis of Ensembles of Topological Descriptors.*
Advisor: *Julien Tierny.*

2018 – 2020
Master's Degree in Computer Science
"*Machine Learning for Data Science*" *track of Paris Descartes University.*
Rank: *1 / 38 (S4) ; 1 / 37 (S3) ; 1 / 33 (S2) and 3 / 33 (S1)*

2016 – 2018
Bachelor's Degree in Computer Science
Toulouse Paul Sabatier University.
Rank: *4 / 152*

2014 – 2016
DUT GEII (Electrical and Computer Science Engineering)
Toulouse Paul Sabatier University.

Awards

2023

- **Best Paper Honorable Mention Award** at IEEE VIS 2023
  For the paper: "Merge Tree Geodesics and Barycenters with Path Mappings"
- **Best Paper and Presentation Award** at CORESA 2023
  For the talk: "Analyse en Géodésiques Principales d'Arbres de Fusion (et de Diagrammes de Persistance)"

Research

Supervision

Ph.D. Student

- Michelle Ditz (Jun. 2026 – Ongoing), co-supervised with Christoph Garth

Master Student

- Mirko Andre Hoehn (Jul. 2026 – Apr. 2027)
  Title: *Wasserstein Neural Network for Persistence Diagrams: Applications to Machine Learning Datasets.*
- Michelle Ditz (Sep. 2025 – Feb. 2026)
  Title: *Wasserstein Branch Mapping Distance for Merge Trees through Complete Branch Decompositions.*

Bachelor Student

- Bastian Schmitt (Jun. 2026 – Jan. 2027)
  Title: *Quantitative Assessment of Shape Similarity Techniques for Discrete Surface Representations.*

Thesis

2023

- **Analysis of Ensembles of Topological Descriptors**
  Mathieu Pont
  Ph.D. thesis in Computer Science
  Committee: *Gabriel Peyré (President), David Coeurjolly (Reviewer), Vijay Natarajan (Reviewer), Elsa Cazelles (Examiner), Stanley Durrleman (Examiner), Roland Kwitt (Examiner), Katharine Turner (Examiner), Julien Tierny (Advisor)*

Publications

2026

- **Operational Abstractions of Neural Network Concepts via Topological Representations**
  Mathieu Pont and Christoph Garth
  Submitted to ICLR.
- **Wasserstein Branch Mapping Distances for Merge Trees**
  Michelle Ditz, Mathieu Pont, Florian Wetzels and Christoph Garth
  To be submitted to PacificVIS the 09/11/2026.
- **Interactive Temporal Merge Tree Maps through Edit Mappings**
  Michael Will, Florian Wetzels, Mathieu Pont, Jonas Lukasczyk and Christoph Garth
  Submitted to PacificVis TVCG Track.

2025

- **Region-Aware Wasserstein Distances of Persistence Diagrams and Merge Trees**
  Mathieu Pont and Christoph Garth
  IEEE Transactions on Visualization and Computer Graphics.
  Accepted, to be presented at IEEE VIS 2026.
- **A Taxonomy for Feature Tracking in Time-Varying Piecewise-Linear Scalar Fields**
  Jonas Lukasczyk, Emma Nilsson, Florian Wetzels, Andrea Schnorr, Michael Will, Mathieu Pont, Ingrid Hotz, Ross Maciejewski, Julien Tierny and Christoph Garth
  IEEE Transactions on Visualization and Computer Graphics. Accepted.

2024

- **A Practical Solver for Scalar Data Topological Simplification**
  Mohamed Kissi, Mathieu Pont, Joshua A. Levine and Julien Tierny
  IEEE Transactions on Visualization and Computer Graphics.
  Proc. of IEEE VIS 2024.

2023

- **Wasserstein Auto-Encoders of Merge Trees (and Persistence Diagrams)**
  Mathieu Pont and Julien Tierny
  IEEE Transactions on Visualization and Computer Graphics.
  Presented at IEEE VIS 2024.
- **Merge Tree Geodesics and Barycenters with Path Mappings**
  Florian Wetzels, Mathieu Pont, Julien Tierny and Christoph Garth
  IEEE Transactions on Visualization and Computer Graphics.
  Proc. of IEEE VIS 2023.
  Best Paper Honorable Mention Award

2022

- **Principal Geodesic Analysis of Merge Trees (and Persistence Diagrams)**
  Mathieu Pont, Jules Vidal and Julien Tierny
  IEEE Transactions on Visualization and Computer Graphics.
  Presented at IEEE VIS 2023.

2021

- **Wasserstein Distances, Geodesics and Barycenters of Merge Trees**
  Mathieu Pont, Jules Vidal, Julie Delon and Julien Tierny
  IEEE Transactions on Visualization and Computer Graphics.
  Proc. of IEEE VIS 2021.

Technical Reports

2025

- **Hands-On TTK Tutorial: Solving Practical Problems in Small Groups**
  Jonas Lukasczyk, Christoph Garth, Michael Will, Mathieu Pont and Julien Tierny
  IEEE VIS Tutorials 2025.

2023

- **A Hands-on TTK Tutorial for Absolute Beginners**
  Christoph Garth, Robin Maack, Mathieu Pont, Julien Tierny, Bei Wang, Florian Wetzels, Michael Will
  IEEE VIS Tutorials 2023.

2022

- **Topological Analysis of Ensemble Scalar Data with TTK, A Sequel**
  Christoph Garth, Charles Gueunet, Pierre Guillou, Federico Iuricich, Joshua Levine, Jonas Lukasczyk, Mathieu Pont, Julien Tierny, Jules Vidal, Bei Wang and Florian Wetzels
  IEEE VIS Tutorials 2022.

Professional Service

Program Committee

- **IEEE VIS Short Papers**
  2025, 2026

Reviewer

- **IEEE VIS**
  2026
- **EuroVis**
  2026
- **La Matematica**
  2024
- **IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)**
  2023

Talks

2026
- **Analysis of Ensembles of Topological Descriptors**
  Jul. 2nd, ERC TORI Workshop Invited Talk

2025
- **Tutorial: Merge Tree Computation and Feature Tracking in TTK**
  Nov. 3rd, IEEE VIS

2024
- **Poster: Wasserstein Auto-Encoders of Merge Trees (and Persistence Diagrams)**
  Nov. 21st, CORESA
- **Wasserstein Auto-Encoders of Merge Trees (and Persistence Diagrams)**
  Oct. 17th, IEEE VIS
- **Variability Analysis of Ensembles of Topological Descriptors**
  Jul. 1st, RPTU Kaiserslautern-Landau Invited Talk
- **Auto-Encodeurs de Wasserstein d'Arbres de Fusion (et de Diagrammes de Persistance)**
  Jun. 18th, Journée Visu
- **Auto-Encodeurs de Wasserstein d'Arbres de Fusion (et de Diagrammes de Persistance)**
  May 30th, Journée APR

2023
- **Analysis of Ensembles of Topological Descriptors**
  Dec. 1st, Ph.D. Defense
- **Principal Geodesic Analysis of Merge Trees (and Persistence Diagrams)**
  Oct. 26th, IEEE VIS
- **Tutorial: Wasserstein Distances between Persistence Diagrams in TTK**
  Oct. 22nd, IEEE VIS
- **Principal Geodesic Analysis of Merge Trees (and Persistence Diagrams)**
  Oct. 16th, Pre-VIS Day
- **Analyse en Géodésiques Principales d'Arbres de Fusion (et de Diagrammes de Persistance)**
  Jun. 23rd, Journée APR
- **Analyse en Géodésiques Principales d'Arbres de Fusion (et de Diagrammes de Persistance)**
  Jun. 22nd, Journée Visu
- **Analyse en Géodésiques Principales d'Arbres de Fusion (et de Diagrammes de Persistance)**
  Jun. 8th, CORESA – Best Paper and Presentation Award!

2022
- **Distances de Wasserstein, Géodésiques et Barycentres d'Arbres de Fusion**
  Nov. 25th, JFIG
- **Tutorial: Wasserstein Distances, Barycenters and Clusters of Merge Trees in TTK**
  Oct. 17th, IEEE VIS, Recorded Talk
- **Distances de Wasserstein, Géodésiques et Barycentres d'Arbres de Fusion**
  Jun. 28th, Journée Visu

2021
- **Wasserstein Distances, Geodesics and Barycenters of Merge Trees**
  Oct. 28th, IEEE VIS, Recorded Talk

Teaching Experience

2024 – 2025
- **RPTU Kaiserslautern-Landau – Scientific Visualization**
  $\sim 2h$ *in Master 2 using C++ and Paraview*

2022 – 2023

- **Sorbonne Université – Introduction to Programming 1**
  $\sim 40h$ *in Bachelor 1 using Python*
- **Sorbonne Université – Data Structures**
  $\sim 20h$ *in Bachelor 2 using C*

2021 – 2022

- **Sorbonne Université – Introduction to Programming 1**
  $\sim 40h$ *in Bachelor 1 using Python*
- **Sorbonne Université – Introduction to Scientific Visualization**
  $\sim 20h$ *in Master 2 using C++ and ParaView*

2020 – 2021

- **Sorbonne Université – Introduction to Scientific Visualization**
  $\sim 20h$ *in Master 2 using C++ and ParaView*
- **Sorbonne Université – Introduction to Programming 2**
  $\sim 40h$ *in Bachelor 1 using C*